Object Oriented Programming (CST-004) - Unit 3

Unit 3 - Inheritance & Polymorphism - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: inheritance and polymorphism - deriving new classes from old ones and making the right function run at runtime.

Key ideas: Inheritance lets a derived class reuse a base class. Types: **single** (one base, one derived), **multilevel** (A → B → C), **multiple** (many bases), **hierarchical** (one base, many derived), **hybrid** (a mix). Access control: public inheritance keeps public members public and protected members protected; private inheritance makes everything private in the derived class. **Polymorphism:** declare a base function **virtual** and a base pointer will call the derived version at runtime (dynamic binding). A **pure virtual** function (= 0) makes the class **abstract** - it cannot be instantiated. In **multiple inheritance** the diamond problem (two copies of a grandparent) is fixed with a **virtual base class**.

Exam pointers: output prediction with virtual vs non-virtual functions is the repeated question; also "types of inheritance with diagrams" and "what is an abstract class / virtual base class". Constructor order in inheritance (base first, then derived) is a favourite 2-marker.

Extra note - RTTI & Casting - made simple:

RTTI (Run-Time Type Identification) = finding out, **while the program is running**, what type an object actually is.

> *Imagine a box labelled "fruit" - inside could be an apple or a mango. RTTI is opening the box to see what is really inside.*

- **typeid(object)** - tells the object's type. Example: `typeid(x).name()`
- **dynamic_cast** - safely converts a base pointer to a derived one. On a wrong conversion it returns **nullptr** (no crash).
- **static_cast** - normal conversion at compile time (like int to float).
- **const_cast** - adds or removes const. **reinterpret_cast** - risky; just re-reads the bits.

Exam tip: dynamic_cast works only when the base class has at least one **virtual function** (a polymorphic class).